SYED RIZWAN UDDIN

San Jose, CA | syed.rizwanuddin@sjsu.edu | LinkedIn | GitHub

SKILLS

Programming & Scripting: Python, C++, Java, JavaScript, SQL, Object-Oriented Programming

Systems & Tools: Linux/Unix, Docker, Docker Compose, Git, Debugging, Automation, Pytest, PostgreSQL, Redis/RQ

Data & AI: NumPy, Pandas, Jupyter, LLM Agents, RAG Pipelines, Sentence Transformers

EDUCATION

San Jose State University **May 2028**

B.S., Computer Science and Linguistics — San Jose, CA

Coursework: Data Structures & Algorithms, Calculus II

GPA: 3.50/4.00

EXPERIENCE

Computational Bioinformatics Research — San Jose State University **Mar 2025 – Jan 2026**

San Jose, CA

- Developed a Python/pandas pipeline (Jupyter) that validates riboswitch CSVs and automatically flags the highest- and lowest-complex-energy interactions across 20+ negative viral targets.
- Analyzed 500+ riboswitch sequences (1,000+ data points) in a Linux VM with ViennaRNA energy, k-mer features, regression and structure plots, comparing functional vs. defective Coxsackie B3 designs for ML-based prediction.

AI Agent Development Intern — NeuralSeek **Aug 2025 – Nov 2025**

Remote | Certification Program

- Set up LLM-powered agents with RAG pipelines and 3+ tool-calling workflows (search, retrieval, API calls) to answer natural-language queries over 3+ data sources.
- Shipped a Japanese-language tutor agent covering all 3 writing systems in under 1 week.

PROJECTS

NullStack — Self-Hosted Code Execution Platform GitHub **Mar 2026**

Python, SQL, Java, Flask, Docker, Docker Compose, Streamlit, Ollama

- Built a self-hosted platform (Streamlit frontend, Flask REST API) that runs untrusted Python, SQL and Java in ephemeral Docker containers with network isolation, 128MB memory caps and 5-second timeouts.
- Integrated a local LLM (Phi-3 Mini) via Ollama for automated code review, deployed with one Docker Compose command.

DriftWatch — Semantic Narrative-Drift Detection Platform GitHub **Aug 2026 – Sep 2026**

Python, JavaScript, FastAPI, PostgreSQL/pgvector, Sentence Transformers, Redis/RQ, Docker Compose, Chart.js

- Built a full-stack NLP web application (FastAPI backend, JavaScript/Chart.js dashboard) that scores how a company's public narrative shifts over time, linking each point to its source article.
- Debugged unexplained drift outliers by tracing them to off-topic source articles, then added a 3-stage relevance gate (entity match, official-domain filter, semantic threshold) to validate inputs before scoring.
- Automated a 10-year search, cutting about 120 monthly windows to at most 12 representative periods (up to 90% fewer planned requests), with retries and per-window failure isolation.
- Containerized the API, worker, PostgreSQL and Redis with Docker Compose; 17 automated Pytest tests.

Centralized Cryptocurrency Exchange — Order-Matching Backend GitHub **Jul 2026**

Python, FastAPI, PostgreSQL, SQLAlchemy, Pydantic, WebSocket, JavaScript

- Implemented an exchange backend from scratch, with a custom matching engine that pairs orders by price-time priority and fills partially against multiple counter-orders, behind a FastAPI REST API.
- Handled concurrent orders with PostgreSQL transactions and row-level locking, separating routes, services, matching engine and repositories into object-oriented layers.

MenuMind — Multi-Tenant Digital Menu Management Platform GitHub **Mar 2026**

Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, Docker Compose, Pytest, Ruff, Mypy

- Owned the backend of a RAG-based restaurant menu chatbot for a 2-person HackHaywards 2026 hackathon team, and led planning by splitting tasks and setting the schedule.
- Designed the solo rebuild as a multi-tenant platform, choosing a modular FastAPI monolith on PostgreSQL over microservices and documenting each architecture decision.